WORKSHEET · SCRATCH (GR 6–8) ADVANCED: ALGORITHMS LAB

Lesson 8: Cryptography: The Caesar Cipher

Module 3 | 45 minutes | Ciphers and modular arithmetic | Grades 6–8 · ages 12–14

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

a. cipher ___ A. Trying every possible key.
b. key ___ B. The remainder after division; wraps numbers around.
c. mod ___ C. The secret value used to encrypt and decrypt.
d. brute force ___ D. A method for encrypting messages.

2. Plan your code

Before you build it: Caesar encrypt: ask for a message, shift each letter by 3 using the alphabet and mod 26, and say the result in out. (Spaces and other characters stay the same.) Write or draw the blocks you will use.

3. Wrap around

With a shift of 3, what does “y” become? Circle one.

a. A. b
b. B. z
c. C. a

4. Plan your code

Before you build it: Caesar decrypt: ask for a secret message and shift each letter by −3 to recover the original. “khoor” should become “hello”. Write or draw the blocks you will use.

5. Explain your thinking

Explain how someone could break a Caesar cipher without knowing the key. Why is modern encryption much safer?

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

- ☐ I can encrypt and decrypt with a Caesar cipher.
- ☐ I can use mod to wrap around.
- ☐ I can explain why it’s easy to break.